

Special matrices, eigenvalues and the tensor product

The last pieces of linear algebra before the quantum rules arrive.

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SESSION 8

Matrices with a special shape

Four families. Two of them run all of quantum mechanics.

Quantum mechanics uses only two kinds of matrix.

Hermitian ones for measurements. Unitary ones for operations.

The dagger: transpose, then conjugate

$$A^\dagger = (A^T)^*$$

$$\begin{pmatrix} 1 & 2i \\ 3 & 4 - i \end{pmatrix}^\dagger = \begin{pmatrix} 1 & 3 \\ -2i & 4 + i \end{pmatrix}$$

Rows become columns, and every i flips sign.

Diagonal matrices

$$D = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$$

Zero everywhere off the diagonal.

$$D \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 2v_1 \\ -3v_2 \end{pmatrix}$$

Each entry is scaled on its own. The identity I is the diagonal matrix with 1s.

Symmetric matrices: $A^T = A$

$$\begin{pmatrix} 1 & 5 \\ 5 & 2 \end{pmatrix}$$

Fold the matrix along its diagonal. The entries that land on each other are equal.

Every diagonal matrix is symmetric.

Hermitian: $M^\dagger = M$

The complex version of symmetric. Mirror across the diagonal, then conjugate.

What $M^\dagger = M$ forces

$$M = \begin{pmatrix} 2 & 1 - i \\ 1 + i & 3 \end{pmatrix}$$

Diagonal entries are real. Each one must equal its own conjugate.

Mirror entries are conjugates. $1 - i$ above, $1 + i$ below.

WORKED EXAMPLE 1

Is M Hermitian? Check it step by step

$$M^T = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix} \longrightarrow (M^T)^* = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$$

$$M^\dagger = M \quad \checkmark$$

WORKED EXAMPLE 2

Is B Hermitian?

$$B = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad B^\dagger = \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$$

$$B^\dagger \neq B$$

So B is symmetric but not Hermitian. The off-diagonal i was never conjugated.

Symmetric and Hermitian differ once i appears

- **Checking only the transpose**

Seeing $B^T = B$ and calling B Hermitian.

Wrong for any matrix with a complex entry off the diagonal.

- **Transpose and conjugate**

Compare M with $M^\dagger$.

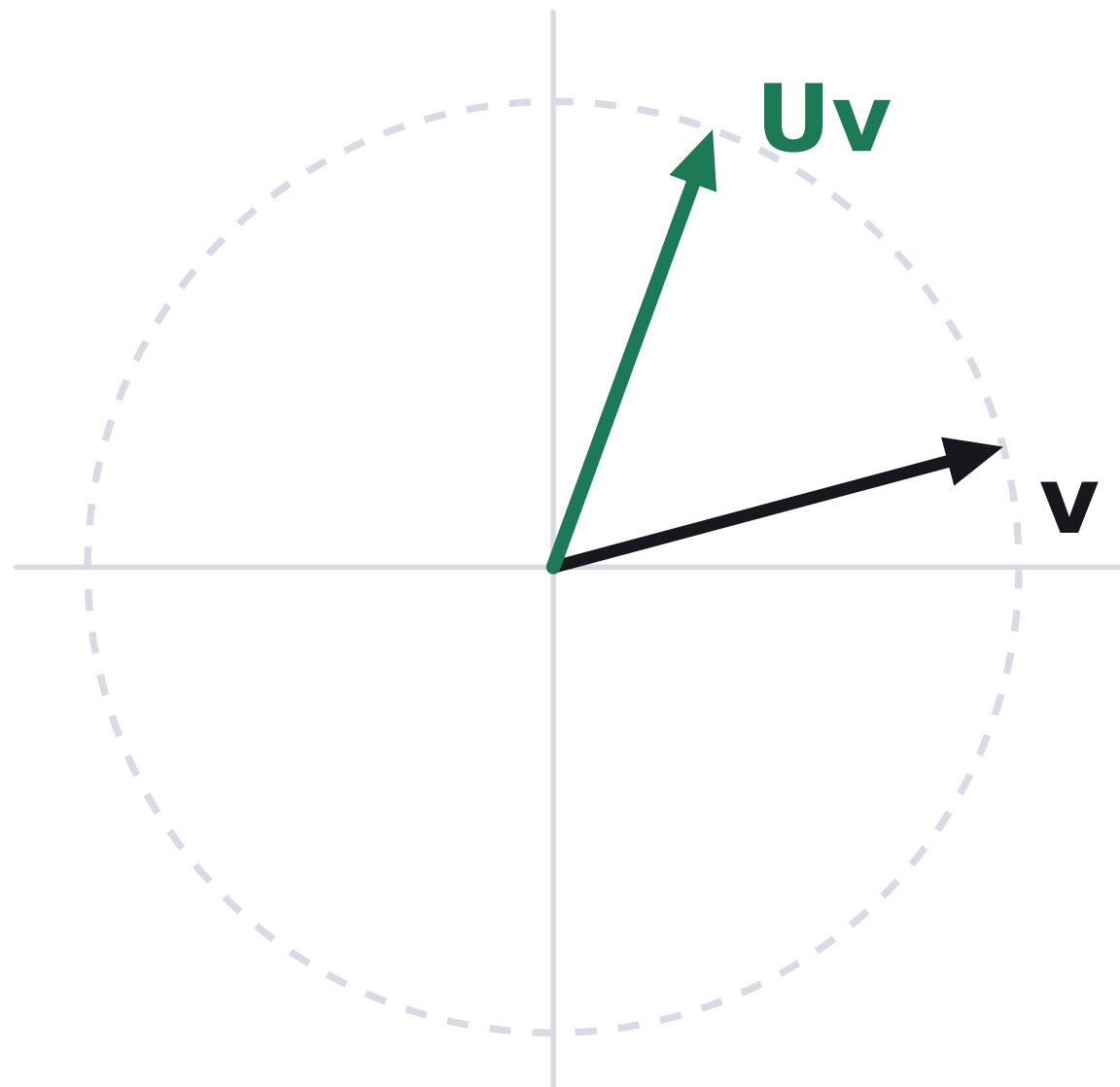
For a **real** matrix the two tests agree, which is why the slip goes unnoticed.

Unitary: $U^\dagger U = I$

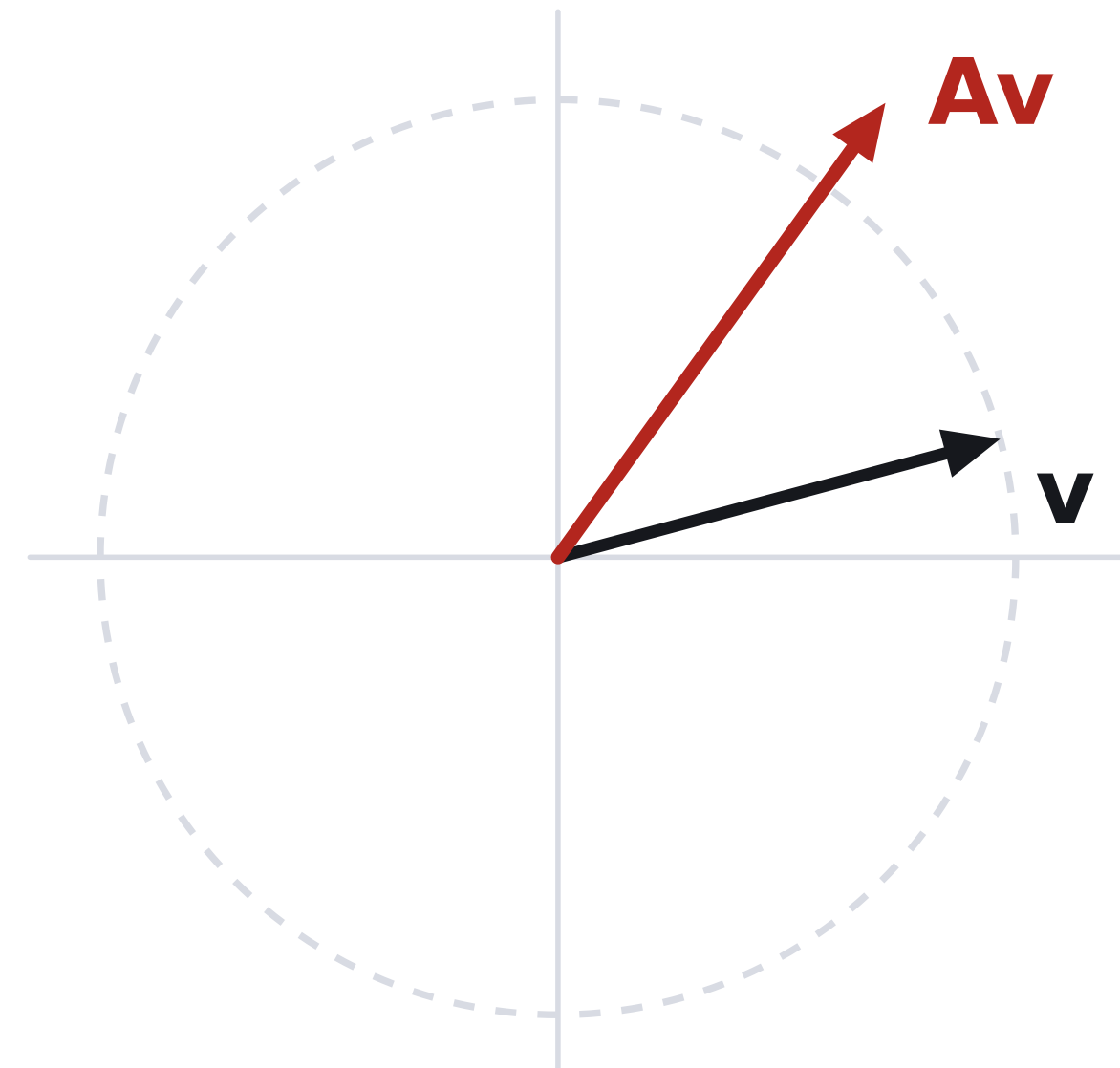
Its dagger is its inverse. Undoing a unitary costs one transpose and one conjugate.

A unitary keeps every length

Unitary U : length kept



Other matrix A : length changed



Lengths and inner products survive

$$\|U\vec{v}\|^2 = (U\vec{v})^\dagger (U\vec{v}) = \vec{v}^\dagger U^\dagger U \vec{v} = \vec{v}^\dagger \vec{v} = \|\vec{v}\|^2$$

$$\langle U\vec{u}, U\vec{v} \rangle = \vec{u}^\dagger U^\dagger U \vec{v} = \langle \vec{u}, \vec{v} \rangle$$

The only rule used: $(AB)^\dagger = B^\dagger A^\dagger$.

Columns orthonormal means unitary

$$(U^\dagger U)_{jk} = \langle \vec{u}_j, \vec{u}_k \rangle, \quad \vec{u}_j = \text{column } j \text{ of } U$$

So $U^\dagger U = I$ says two things: each column has norm 1, and different columns are orthogonal.

WORKED EXAMPLE 3

Is this U unitary? Test the columns

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \quad \vec{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \vec{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\|\vec{u}_1\|^2 = \frac{1}{2} (1 + |i|^2) = 1 \quad \langle \vec{u}_1, \vec{u}_2 \rangle = \frac{1}{2} (1 + (i)^*(-i)) = \frac{1}{2} (1 - 1) = 0$$

Same for $\vec{u}_2$. Orthonormal columns, so U is unitary.

TRY THIS NOW · 4 MINUTES

Hermitian, unitary, both, or neither?

$$(a) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (b) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad (d) \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

ANSWERS

Two are both, one is neither

	HERMITIAN	UNITARY	REASON
(a)	yes	yes	real symmetric; columns orthonormal
(b)	yes	yes	$-i$ and i are conjugates
(c)	no	no	columns $(1, 0)$ and $(1, 1)$ overlap: inner product 1
(d)	no	yes	diagonal i is not real

You will meet (a), (b) and (d) again in session 13, as the gates X , Y and S .

The four families

FAMILY	RULE	ROLE IN THIS COURSE
diagonal	zero off the diagonal	eigenvalues sit on show
symmetric	$A^T = A$	real special case
Hermitian	$M^\dagger = M$	measurements (session 15)
unitary	$U^\dagger U = I$	operations (session 13)

Test a matrix in two lines

```
import numpy as np
U = np.array([[1, 1], [1j, -1j]]) / np.sqrt(2)
Ud = U.conj().T           # the dagger
print(np.allclose(Ud @ U, np.eye(2))) # unitary? True
print(np.allclose(Ud, U))   # Hermitian? False
```

In Python the imaginary unit is written `1j`.

What to do before next session

- Redo worked example 3 without looking.
- Check (a) to (d) from the exercise in NumPy.
- Revise how to solve a quadratic equation. Session 9 needs it.

SESSION 9 · BEYOND THE SYLLABUS · NEEDED IN UNIT 3

Eigenvalues, eigenvectors and the tensor product

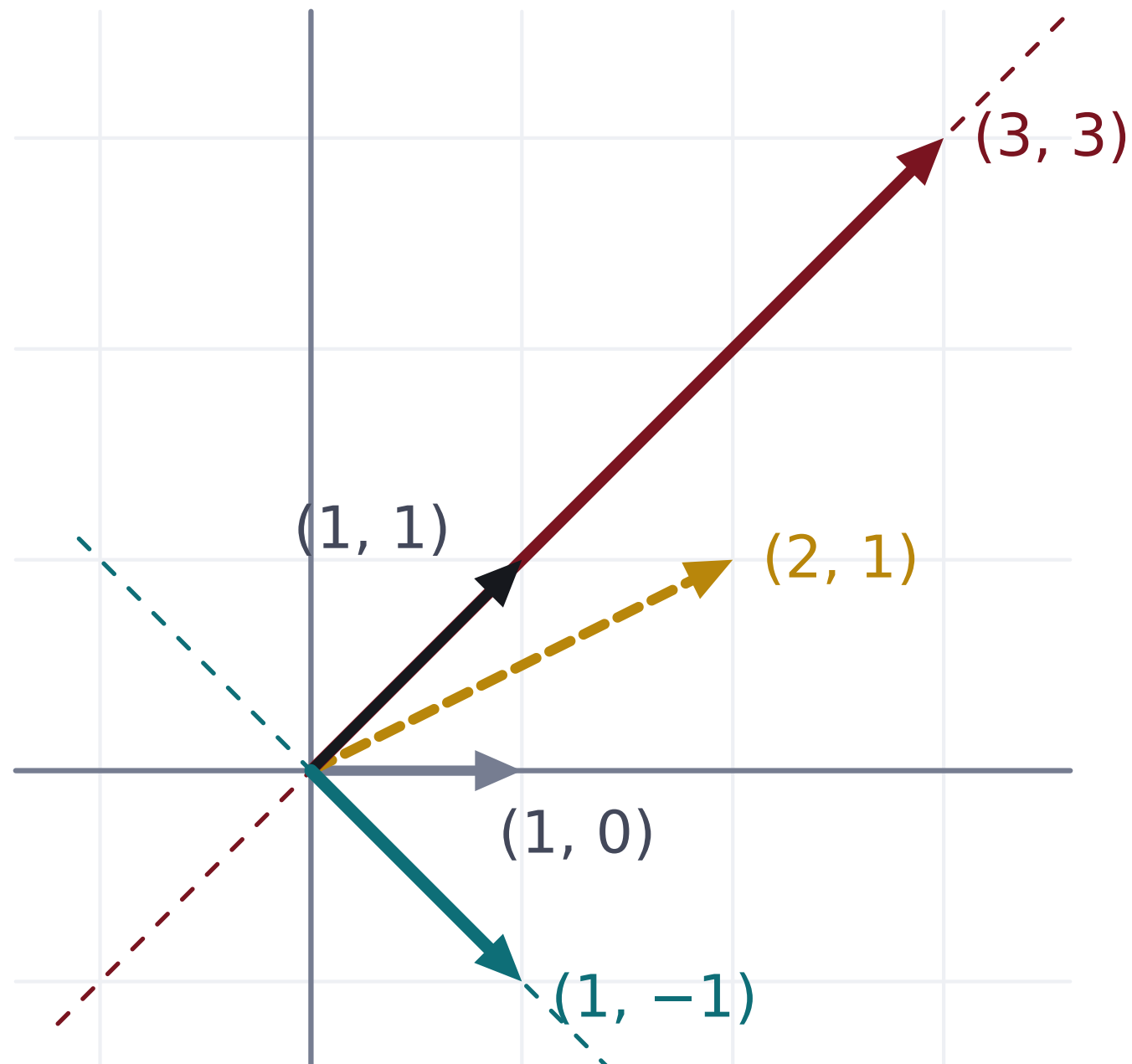
Two tools the syllabus leaves out and Unit 3 cannot do without.

$$A\vec{v} = \lambda\vec{v}$$

An **eigenvector** $\vec{v}$ is only stretched by A . The stretch factor λ is its **eigenvalue**.

PICTURE IT

Most vectors turn. Eigenvectors stay on their line.



What M does to three vectors

■ $(1, 0) \rightarrow (2, 1)$
direction changes

■ $(1, 1) \rightarrow (3, 3)$
same line, $\lambda = 3$

■ $(1, -1) \rightarrow (1, -1)$
same line, $\lambda = 1$

WORKED EXAMPLE 1

Checking a claimed eigenvector

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
$$A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

First: same vector back, so $\lambda = 1$.

Second: $\begin{pmatrix} -1 \\ 1 \end{pmatrix} = -1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, so $\lambda = -1$.

To check, multiply and compare. No formula needed.

The characteristic equation

$$(A - \lambda I)\vec{v} = \vec{0} \text{ with } \vec{v} \neq \vec{0} \implies \det(A - \lambda I) = 0$$

$$\det \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} = (a - \lambda)(d - \lambda) - bc = 0$$

For a 2×2 matrix, det is $ad - bc$. A quadratic in λ gives two eigenvalues.

Eigenvalues of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

$$(2 - \lambda)^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0 \quad \implies \quad (\lambda - 1)(\lambda - 3) = 0$$

$$\lambda = 1 \quad \text{or} \quad \lambda = 3$$

One eigenvector for each eigenvalue

$$\lambda = 3 : \quad \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \vec{0} \implies v_1 = v_2 \implies \vec{v} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda = 1 : \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \vec{0} \implies v_2 = -v_1 \implies \vec{v} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Subtract λ from the diagonal, then read off the relation between v_1 and v_2 .

Back to M from session 8

$$M = \begin{pmatrix} 2 & 1 - i \\ 1 + i & 3 \end{pmatrix} : \quad (2 - \lambda)(3 - \lambda) - (1 - i)(1 + i) = 0$$

$$\lambda^2 - 5\lambda + 4 = 0 \quad \implies \quad \lambda = 1, 4$$

Complex entries, yet both eigenvalues are real. That is no accident.

And they are orthogonal

$$\lambda = 1 : \vec{a} = \begin{pmatrix} -1 + i \\ 1 \end{pmatrix} \qquad \lambda = 4 : \vec{b} = \begin{pmatrix} 1 - i \\ 2 \end{pmatrix}$$

$$\langle \vec{a}, \vec{b} \rangle = (-1 + i)^*(1 - i) + 1 \cdot 2 = (-1 - i)(1 - i) + 2 = -2 + 2 = 0$$

Normalise before use: divide by $\sqrt{3}$ and $\sqrt{6}$.

Hermitian $\rightarrow$ real eigenvalues, orthogonal eigenvectors

This is why every measurement outcome in session 15 is a real number.

"My eigenvector differs from the book's"

- **Panicking**

You found $\begin{pmatrix} -1+i \\ 1 \end{pmatrix}$. The book says $\begin{pmatrix} 1 \\ \frac{-1-i}{2} \end{pmatrix}$.

- **Checking the ratio**

Multiply yours by $\frac{-1-i}{2}$ and you get the book's.

Any nonzero multiple of an eigenvector is the same eigenvector. The zero vector never counts.

TRY THIS NOW · 5 MINUTES

Eigenvalues and eigenvectors of

$$T = \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$$

First check T is Hermitian. Then find both eigenvalues and one eigenvector for each.

ANSWER

$$\lambda = 0 \text{ and } \lambda = 2$$

$$(1 - \lambda)^2 - (-i)(i) = (1 - \lambda)^2 - 1 = 0$$

$$T \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} 2 \\ 2i \end{pmatrix} = 2 \begin{pmatrix} 1 \\ i \end{pmatrix} \quad T \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Real eigenvalues, and $(1, i)$ is orthogonal to $(1, -i)$.

Each entry of the first multiplies the whole second

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \otimes \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_1 \\ a_2 b_2 \end{pmatrix}$$

Two qubits need four entries. The tensor product builds them.

WORKED EXAMPLE 4

Two tensor products

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \otimes \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \cdot 3 \\ 1 \cdot 4 \\ 2 \cdot 3 \\ 2 \cdot 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 6 \\ 8 \end{pmatrix} \qquad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

Dimensions multiply: $\mathbb{C}^m \otimes \mathbb{C}^n = \mathbb{C}^{mn}$. Norms multiply too.

The Kronecker product

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix}$$

Same rule as vectors: each entry of A multiplies a whole copy of B . Two 2×2 matrices give a 4×4 .

WORKED EXAMPLE 5

A Kronecker product, block by block

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 \cdot B & 2 \cdot B \\ 0 \cdot B & 1 \cdot B \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Useful rule: $(A \otimes B)(\vec{a} \otimes \vec{b}) = (A\vec{a}) \otimes (B\vec{b})$.

Order matters

- Assuming it commutes

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \otimes \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 6 \\ 8 \end{pmatrix}$$

- Keeping the order

$$\begin{pmatrix} 3 \\ 4 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 4 \\ 8 \end{pmatrix}$$

Not every $\mathbb{C}^4$ vector is a product

$$\vec{w} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_1 \\ a_2 b_2 \end{pmatrix}$$

Entries 1 and 4 need a_1, b_1, a_2, b_2 all nonzero. Entry 2 needs $a_1 b_2 = 0$. Contradiction. Session 14 calls this **entanglement**.

TRY THIS NOW · 4 MINUTES

Three quick ones

1. Compute $\begin{pmatrix} 1 \\ i \end{pmatrix} \otimes \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.
2. Is $(1, 1, 1, 1)$ a tensor product of two vectors in $\mathbb{C}^2$?
3. How many entries does a vector in $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$ have?

ANSWERS

Check yours

1. $(2, -1, 2i, -i)$

2. Yes: $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

3. $2 \times 2 \times 2 = 8$

```
np.kron([1, 1j], [2, -1])    # array([2, -1, 2j, -1j])
```

End of Unit 2

- **Assignment Part B** (linear algebra) is due at the end of week 10.
- **Class exam** on Evolveus, Units 1 and 2, in the test slot after this week.
- Revise the inner product. Session 10 rewrites it as $\langle \phi | \psi \rangle$.